Uniqueness of positive harmonic profiles outside convex cones for general stable operators

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Abstract

We prove uniqueness, up to a positive factor, of positive harmonic functions outside a proper closed convex cone with nonempty interior for every nondegenerate symmetric stable operator. The spectral measure may be singular or supported on finitely many directions. Each profile is assumed continuous, zero on the cone, and of polynomial growth strictly below the operator's order. The proof controls the entire growth norm by one interior value. This follows by normalizing at a maximizing scale and using a lower barrier whose exponent is smaller than the chosen growth weight. A comparison of the resulting normalized profiles then yields proportionality.

1 Introduction

Positive harmonic functions that vanish on a cone are natural objects in the study of nonlocal elliptic equations and their blow-up limits. For symmetric stable operators, their behavior reflects both the cone geometry and the directions in which jumps are allowed. A finite spectral measure describes those directions, and it can be concentrated on a few of them. The associated interior and boundary regularity theory was developed by Ros-Oton and Serra [2]. Their half-space Liouville theorem [2, Theorem 4.1] classifies weak solutions of sub-$2s$ polynomial growth in a half-space.

For more general cones, proportionality of positive profiles follows from boundary Harnack inequalities in the pointwise elliptic setting; see [1, Proposition 4.4.5]. Such an approach does not apply to all stable operators, since even interior Harnack inequalities can fail for singular spectral measures. A related uniqueness question is stated in [1, Open question 4.4, p. 302]. Ros-Oton and Weidner [3, Theorem 4.6 and Corollary 4.7] establish comparison of positive cone profiles under an angular L^r condition, $r > n/(2s)$, and directional monotonicity assumptions. Their argument controls nonlocal tails before comparing functions near the cone.

Here we establish proportionality for all nondegenerate finite spectral measures, including atomic ones, for profiles with an explicit growth exponent below the order of the operator. The geometric assumption is that the closed convex cone is proper and has nonempty interior. These conditions specify the version of the uniqueness problem proved below; no conclusion about other growth classes or cones with empty interior is asserted.

Let $n \geq 2$, $0 < s < 1$, and let μ be a finite even nonnegative Borel measure on S^{n-1} such that, for some $\lambda, \Lambda > 0$,

$$\mu(S^{n-1}) \leq \Lambda, \qquad \inf_{|e|=1} \int_{S^{n-1}} |e \cdot \theta|^{2s} \, \mu(d\theta) \geq \lambda. \tag{1.1}$$

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The second condition excludes measures supported in a hyperplane. Define the positive operator

$$Lu(x) = \frac{1}{2}\int_{S^{n-1}}\int_0^\infty \big(2u(x) - u(x+r\theta) - u(x-r\theta)\big)\frac{dr}{r^{1+2s}}\,\mu(d\theta). \tag{1.2}$$

For continuous solutions the equation is understood in the viscosity sense. The growth condition in the following theorem makes its tails integrable.

Theorem 1.1. *Let L satisfy (1.1)–(1.2), and let $\Sigma \subsetneq \mathbb{R}^n$ be a closed convex cone with nonempty interior. Suppose $w_1, w_2 \in C(\mathbb{R}^n)$ are zero on Σ, strictly positive on $\mathbb{R}^n \setminus \Sigma$, and satisfy*

$$Lw_i = 0 \quad \text{in } \mathbb{R}^n \setminus \Sigma$$

in the viscosity sense. If, for each $i = 1, 2$, there are finite C_i and $\beta_i < 2s$ with

$$|w_i(x)| \le C_i(1 + |x|)^{\beta_i} \qquad (x \in \mathbb{R}^n),$$

then $w_1 = cw_2$ on $\mathbb{R}^n$ for some $c > 0$.

The functions need not be homogeneous or monotone. The constants used in the proof concern a fixed operator and cone; uniformity over all such operators and cones is unnecessary for the proportionality conclusion.

The difficulty is to obtain comparison constants independent of a profile's initial tail bound. Write $\alpha = 2s$. A ball barrier has some exponent $q < \alpha$, and one can choose a growth weight p above q and both actual profile exponents, while retaining $p < \alpha$. Normalize a profile at a scale maximizing its weighted supremum. A compactness argument and supporting balls then give a distance lower bound for that normalized function. Evaluation at a fixed interior point recovers the factor $R^{q-p} \le 1$, and hence controls the whole growth norm by that single value. This controls tails along every direction charged by μ.

The required compactness includes the cone boundary. We obtain uniform boundary smallness from the stopped stable process: starting points that approach the cone have expected exit times tending to zero. With the growth bound in hand, an annular comparison in the spirit of [3] works using positive spectral mass in outward directions. Scaling makes the comparison global, and an extremal nonnegative difference proves uniqueness.

Section 2 establishes the solution-concept and stopping facts. Section 3 proves boundary compactness and growth control, and Section 4 proves proportionality. Throughout, $B_R = B_R(0)$ is an open Euclidean ball, suprema are pointwise, and the generator of the associated stable process is $-L$.

2 Analytic facts and stopped harmonic functions

Fix the operator and cone in Theorem 1.1, and put $\alpha = 2s$ and $D = \mathbb{R}^n \setminus \Sigma$. For $0 \le p < \alpha$, let

$$\mathcal{B}_p = \left\{u : \sup_{x\in\mathbb{R}^n} \frac{|u(x)|}{(1+|x|)^p} < \infty\right\}, \qquad \|u\|_{\mathcal{B}_p} = \sup_x \frac{|u(x)|}{(1+|x|)^p}.$$

The weight is equivalent to $1 + |x|^p$, the convention in [1]. A distributional equation $Lu = f$ means $\int uL\varphi = \int f\varphi$ for every $\varphi \in C_c^\infty$ supported in the domain. Viscosity inequalities use a local C^2 touching test, patched to the original function outside the touching ball. Upper tests give $L \le f$, and lower tests give $L \ge f$.

The symmetric Levy measure associated with our normalization is

$$\nu(dz) = r^{-1-\alpha}\, dr\, \mu(d\theta), \qquad z = r\theta, \quad r > 0. \tag{2.1}$$

This formula denotes a pushforward measure and permits atoms in μ. For a smooth function with growth below α,

$$Lu(x) = \int \big(u(x) - u(x+z) + \mathbf{1}_{|z|<1} \nabla u(x) \cdot z\big)\, \nu(dz). \tag{2.2}$$

Symmetry cancels the odd term and gives exactly (1.2).

Here are the general-measure analytic results we use. Their applicability can be checked directly. Write $M_\mu = \mu(S^{n-1})$, and set $\delta = (\lambda/(2\Lambda))^{1/\alpha}$. For every unit e, the contribution of $|e \cdot \theta| < \delta$ to its α-moment is at most $\lambda/2$. Consequently

$$\mu\{|e \cdot \theta| \geq \delta\} \geq \lambda/2, \qquad \int |e \cdot \theta|^2\, \mu(d\theta) \geq \delta^2 \lambda/2. \tag{2.3}$$

Radial integration gives

$$r^\alpha \nu(B_{2r} \setminus B_r) = \frac{M_\mu}{\alpha}(1 - 2^{-\alpha}), \tag{2.4}$$

$$r^{\alpha-2} \int_{B_r} |e \cdot z|^2 \nu(dz) \geq \frac{\delta^2 \lambda}{2(2-\alpha)}. \tag{2.5}$$

Thus L belongs to the general stable class of [1, Definitions 2.1.18 and 2.1.21]. The class constants may differ from λ, Λ, which causes no difficulty for a fixed operator. In particular, that source permits the measure in (2.1).

Lemma 2.1 (General stable analytic results)**.** *For the fixed operator L, the following statements hold.*

1. *If a continuous $u \in \mathcal{B}_p$ is distributionally harmonic in a ball, its convolutions with smooth compactly supported mollifiers are smooth and strongly harmonic in smaller balls. Interior C^s estimates there are controlled by $\|u\|_{\mathcal{B}_p}$.*

2. *On a ball, continuous exterior data in $\mathcal{B}_p$ have a unique continuous distributionally harmonic extension in the same growth class.*

3. *On a bounded connected domain, a globally nonnegative continuous distributionally harmonic function in $\mathcal{B}_p$ is either positive throughout that domain or identically zero there.*

4. *There are $q \in (s, \alpha)$ and $\eta \in (0, 1)$ such that*

$$\phi(x) = (1 - |x|^2)_+^q, \qquad L\phi \leq -1 \quad \text{in } B_1 \setminus \overline{B}_{1-\eta}. \tag{2.6}$$

The respective sources are [1, Lemma 2.2.30 and Theorem 2.4.3], [1, Theorem 2.6.12], [1, Theorem 2.4.15(b)], and [1, Lemma B.1.4]. For the interior estimate choose the source's Lebesgue exponent $2n/\alpha$; the resulting Holder exponent is $\alpha - \alpha/2 = s \in (0, 1)$. This also covers $s = 1/2$. The Dirichlet existence result is used only on smooth balls with continuous data; no regularity of the cone boundary is needed for that use.

We will also use stability of the distributional equation under local uniform convergence with one common $\mathcal{B}_p$ bound. To see the relevant tail estimate for a test $\varphi \in C_c^\infty$, take R larger than

twice its support radius. A jump from $|x| > R$ into that support has length at least $R/2$. Changing the order of absolutely integrable integrals gives

$$\int_{|x|>R} (1+|x|)^p |L\varphi(x)|\,dx \le C_\varphi \int_{R/2}^{\infty} (1+r)^p r^{-1-\alpha}\,dr \longrightarrow 0. \tag{2.7}$$

On bounded sets $L\varphi$ is bounded by its local second derivatives and the finite tail mass. This proves both that the distributional pairing is defined and that the stated stability holds for this fixed L.

Lemma 2.2 (Comparison and solution concepts)**.** *Let Ω be bounded and open. Suppose $a, b \in C(\mathbb{R}^n) \cap \mathcal{B}_p$, $La \le 0 \le Lb$ in Ω in the viscosity sense, and $a \le b$ on Ω^c. Then $a \le b$ on $\mathbb{R}^n$. Moreover, a continuous function of growth below α is viscosity harmonic in an open set if and only if it is distributionally harmonic there.*

Proof. Choose $0 \le \psi \le 1$ smooth and compactly supported, with $\psi = 1$ on a ball containing $\overline{\Omega}$. Every jump contribution to $L\psi$ in Ω is nonnegative, and sufficiently long jumps give $L\psi \ge c > 0$ there. For a large radial cutoff χ_R, equal to one on B_R and zero outside B_{2R}, the separated tails give

$$L(\chi_R a) \le e_R, \qquad L(\chi_R b) \ge -e_R, \qquad e_R \le CR^{p-\alpha} \longrightarrow 0.$$

These are viscosity inequalities: near a test point the functions are unchanged, and the changed tail is bounded by $C\int_R^\infty (1+r)^p r^{-1-\alpha}\,dr$. Fix $t > 0$, take $2e_R < tc$, and put $A = \chi_R a$, $B = \chi_R b + t\psi$. They are ordered outside Ω, and $LA \le e_R < tc - e_R \le LB$.

If $\sup(A - B) > 0$, maximize $A(x) - B(y) - |x-y|^2/(2\varepsilon)$ globally. Compact support gives maximizers, and exterior order and continuity place both in Ω for sufficiently small ε. At these points,

$$A(x) - A(x+z) - B(y) + B(y+z) \ge 0 \qquad (z \in \mathbb{R}^n).$$

Use the quadratic upper test for A and lower test for B. Their gradients coincide. Far jumps have nonnegative operator difference; for jumps of length less than r, the difference is bounded below by

$$-\varepsilon^{-1} \int_{|z|<r} |z|^2 \nu(dz) = -\frac{M_\mu}{\varepsilon(2-\alpha)} r^{2-\alpha}.$$

Let $r \downarrow 0$ at fixed ε. This contradicts the strict negative upper bound $2e_R - tc$ for the same operator difference. Thus $A \le B$. On Ω, the cutoff is one, and $t \downarrow 0$ proves comparison.

For a continuous distributional solution, the harmonic mollifications in Lemma 2.1(1) converge locally uniformly with a common sub-α growth bound. The viscosity tests pass to the limit: the patched test controls small jumps, local uniform convergence controls bounded jumps, and polynomial domination controls the tail. Thus this solution is viscosity harmonic. Conversely, solve the distributional Dirichlet problem on a ball compactly contained in the given open set, using the viscosity solution as exterior data. By the preceding argument the extension is viscosity harmonic. Comparison in both directions identifies it with the original function. This proves the converse locally on every such ball, and hence on the open set. □

In particular, linear combinations of our harmonic functions are harmonic. Multiplication by a smooth cutoff equal to one near the tested domain changes the equation only by its explicit tail integral. Addition of a smooth function changes the right-hand side by its classical L-image.

To control boundary values uniformly over profiles, we use the symmetric stable process X with Levy triplet $(0, 0, \nu)$, started at zero. Existence, the generator on C_c^∞, the jump decomposition,

and the strong Markov property are given in [4, Results 7.5, 6.8, 9.7–9.12, and 4.12]. Our sign is $\mathcal{A} = -L$. Indeed its characteristic exponent is

$$\Psi(\xi) = \int (1 - \cos(\xi \cdot z))\nu(dz) = c_\alpha \int |\xi \cdot \theta|^\alpha \mu(d\theta), \quad c_\alpha = \int_0^\infty (1 - \cos r) r^{-1-\alpha}\, dr \in (0, \infty).$$

It follows that X_t has the law $t^{1/\alpha} X_1$. Its semigroup is Feller: for $f \in C_0(\mathbb{R}^n)$, dominated convergence shows that $\mathbb{E} f(x + X_t)$ is continuous and vanishes at infinity, while uniform continuity gives $\|P_t f - f\|_\infty \le \mathbb{E}\omega_f(|X_t|) \to 0$. Thus Dynkin's formula [4, Lemma 12.2] applies to compact smooth tests and finite-mean stopping times.

Lemma 2.3 (Stopped representation)**.** *Let $\Omega \subset D$ be bounded and open, and let $u \in C(\mathbb{R}^n) \cap \mathcal{B}_b$, $0 \le b < \alpha$, be harmonic in D. For $x \in \Omega$ and $\tau_\Omega^x = \inf\{t > 0 : x + X_t \notin \Omega\}$,*

$$u(x) = \mathbb{E} u(x + X_{\tau_\Omega^x}). \tag{2.8}$$

The stopped positions in the proof have a uniformly bounded moment of every fixed order strictly between b and α.

Proof. We give the stopping and integrability details because the spectral measure can be singular. The filtration is the usual completed right-continuous natural filtration. If stopping times $\sigma_j \uparrow \sigma < \infty$ almost surely, the strong Markov property implies, for $a, \varepsilon, M > 0$,

$$\begin{aligned} &\mathbb{P}(|X_\sigma - X_{\sigma_j}| > a,\ \sigma \le M) \\ &\quad \le \mathbb{P}(\sigma - \sigma_j > \varepsilon,\ \sigma \le M) + \mathbb{P}\left(\sup_{0 \le t \le \varepsilon} |X_t| > a \right). \end{aligned}$$

Let $j \to \infty$, then $\varepsilon \downarrow 0$, then $M \to \infty$. This proves convergence in probability. Cadlag paths already give a left limit on paths where the sequence strictly announces σ, so that limit equals X_σ almost surely. Where the times eventually equal σ, convergence is immediate. We have proved the required quasi-left continuity.

For completeness, exits from our bounded open sets are stopping times and their positions belong to the complement. If $\Omega \subset B_A$, put $F = \Omega^c$ and $\sigma_m = \inf\{t > 0 : \operatorname{dist}(x + X_t, F) < 1/m\}$, starting with large m. Hitting this open neighborhood is a stopping time by rational approximation and right continuity. Its position has distance at most $1/m$ from F. Every σ_m is bounded by T_{2A}, the first jump longer than $2A$: a path still in B_A cannot remain there after such a jump. This time is exponential, of rate $M_\mu(2A)^{-\alpha}/\alpha > 0$. The increasing limit σ is finite; quasi-left continuity gives $x + X_\sigma \in F$. Since $\sigma_m \le \tau_\Omega^x \le \sigma$, the limit is the exit time, with the asserted properties.

Choose $p' \in (b, \alpha)$ and a nonnegative smooth $G \asymp (1 + |y|)^{p'}$. For compact smooth cutoffs $G_N = \chi_N G$, $0 \le \chi_N \le 1$, $\chi_N = 1$ on B_N, the generators $\mathcal{A}G_N$ are uniformly bounded on B_A, for $N > 4A$. Small jumps are bounded by local second derivatives times $|z|^2$; large jumps by $C(1+|z|)^{p'}$, which is integrable against the tail of ν. Dynkin and Fatou therefore give, for every stopping time $\sigma \le \tau_\Omega^x$ and finite horizon T,

$$\mathbb{E} G(x + X_{\sigma \wedge T}) \le G(x) + C\mathbb{E} T_{2A} < \infty. \tag{2.9}$$

The same estimate holds for exits from inner subsets of Ω.

Exhaust Ω by $\Omega_j = \{y \in \Omega : \operatorname{dist}(y, \Omega^c) > 1/j\}$, and write τ_j for their exits. Mollify u at a scale below the distance of $\overline{\Omega}_j$ from D^c. The mollification is smooth and harmonic on Ω_j. Apply Dynkin first to its compact smooth cutoffs and $\tau_j \wedge T$. The cutoff generator error tends uniformly to zero,

with bound $CN^{b-\alpha}$. The moment estimate permits removal of the cutoff and the mollifier. If $b = 0$, the function values are bounded. If $b > 0$, put $r = p'/b > 1$; then

$$\sup_{j,T} \mathbb{E}|u(x + X_{\tau_j \wedge T})|^r < \infty, \qquad \mathbb{E}(|Y|\mathbf{1}_{|Y|>H}) \le H^{1-r}\mathbb{E}|Y|^r.$$

This proves uniform integrability, including for the mollifications. Consequently $T \to \infty$ gives $u(x) = \mathbb{E}u(x + X_{\tau_j})$.

Finally $\tau_j \uparrow \tau_\Omega^x$. Otherwise the limiting position, given by quasi-left continuity, would be interior to Ω, whereas each exited inner position has distance at most $1/j$ from Ω^c. Stopped positions converge almost surely; continuity of u and the same uniform integrability give (2.8). □

3 Boundary compactness and growth control

The operator and cone stay fixed throughout this section. Constants may depend on them, on a chosen $p < \alpha$, and on the indicated compact sets. They will be uniform over the functions in each stated family.

Proposition 3.1 (Uniform boundary smallness)**.** *Fix $0 < p < \alpha$ and $K < \infty$. Consider all nonnegative continuous functions, zero on Σ, harmonic in D, and bounded by*

$$u(x) \le K(1 + |x|)^p. \tag{3.1}$$

For every $0 < A < \infty$, these functions tend uniformly to zero as points of $D \cap \overline{B}_A$ approach Σ.

Proof. We first show that X_1 has full support. The support of μ spans $\mathbb{R}^n$ by directional nondegeneracy. Choose independent support directions $\theta_1, \dots, \theta_n$. Write the center of any prescribed ball as $\sum b_j\theta_j$. For nonzero coefficients, small disjoint angular and radial neighborhoods of $b_j\theta_j$ have positive ν-mass; use evenness if $b_j < 0$. Requiring one jump in each bin and no other jump longer than ε has positive probability. Choose the bins so their sum lies in half the prescribed ball, and then take ε below their radial supports. The remaining independent symmetric small-jump martingale has

$$\mathbb{E}|X_1^{\le\varepsilon}|^2 = \int_{|z|\le\varepsilon} |z|^2\nu(dz) = \frac{M_\mu}{2-\alpha}\varepsilon^{2-\alpha} \longrightarrow 0.$$

Chebyshev's inequality gives positive probability that this remaining part lies in the other half-ball. The jump assertions follow from [4, Theorems 9.7, 9.9, 9.12 and Corollary 9.11]. Symmetry makes the intermediate compensating drift zero. This proves full support, including for atomic spectral measures.

Since $\operatorname{int}\Sigma$ is nonempty and invariant under positive dilations, scaling gives

$$\mathbb{P}(X_t \in \operatorname{int}\Sigma) = \mathbb{P}(X_1 \in \operatorname{int}\Sigma) = p_0 > 0 \quad (t > 0).$$

The event of visits to this open cone at arbitrarily small positive times therefore has probability at least p_0. It is a germ event, and has probability one. Indeed a germ event is independent of the increments after every $\varepsilon > 0$. As $\varepsilon \downarrow 0$, right continuity shows independence from every finite collection of path values, hence from the whole path sigma algebra and from itself.

Let $\Omega = D \cap B_{2A+2}$, and couple all starting points by the same process. If $x_k \in D \cap \overline{B}_A$ tends to $y \in \Sigma$, then

$$\tau_\Omega^{x_k} \longrightarrow 0 \quad \text{almost surely}, \qquad \mathbb{E}\tau_\Omega^{x_k} \longrightarrow 0. \tag{3.2}$$

To prove the first assertion, use any of the arbitrarily early times with $X_t \in \operatorname{int}\Sigma$. Convexity of the cone gives $y + \operatorname{int}\Sigma \subset \operatorname{int}\Sigma$, so $x_k + X_t \in \Sigma$ for all sufficiently large k. The second assertion follows by domination with the same T_{4A+4}.

Choose a smooth nonnegative F_A, zero on B_A, dominating $K(1+|x|)^p$ outside B_{2A+2}, with this polynomial growth. Its generator is bounded on B_{2A+2}. At exit, $u = 0$ on Σ and $u \le F_A$ outside the ball. Lemma 2.3 gives

$$0 \le u(x) \le \mathbb{E}F_A(x + X_{\tau_\Omega^x}) \le C_A \mathbb{E}\tau_\Omega^x, \qquad x \in D \cap B_A. \tag{3.3}$$

For the last inequality apply Dynkin to nonnegative compact cutoffs of F_A, whose generators have one common bound on the stopped ball, and then use Fatou. Equations (3.2)–(3.3) and compactness of $\Sigma \cap \overline{B}_A$ prove the assertion. □

Lemma 3.2 (Compactness and positivity)**.** *The family in Proposition 3.1 is precompact for local uniform convergence on $\mathbb{R}^n$. A subfamily normalized by $\sup_{B_1} u = 1$, or by $u(x_*) = 1$ at a fixed $x_* \in D$, has a common positive lower bound on every compact subset of D.*

Proof. Interior Holder estimates from Lemma 2.1, together with the uniform smallness near Σ, give equicontinuity on every closed ball. Arzela–Ascoli and a diagonal extraction yield local uniform compactness. The common growth bound and (2.7) preserve distributional harmonicity. Limits are nonnegative, continuous, and zero on Σ.

The set D is path connected. Choose $e \in \operatorname{int}\Sigma$. For $x \notin \Sigma$, the ray $x - te$, $t \ge 0$, stays outside Σ, since otherwise adding $te \in \Sigma$ would put x in Σ. There is a nonzero functional $\ell \ge 0$ on Σ: project any point outside the cone onto it and take the negative outward normal. Since e is interior, $\ell(e) > 0$. For two given points, their translates by a sufficiently large common $-te$ lie in the half-space $\{\ell < 0\}$. Join them there and along their rays.

A nonzero nonnegative limit is consequently positive throughout D, by the strong maximum principle in Lemma 2.1(3), applied on bounded connected neighborhoods of such paths. Both stated normalizations pass to local uniform limits and exclude the zero function. If a compact lower bound failed, a convergent sequence of points on that compact set and a locally uniformly convergent sequence of profiles would give a nonzero limit with an interior zero, a contradiction. □

We next turn the controlled polynomial bound into one independent of the profile's initial growth constant. Let q, η be as in (2.6).

Lemma 3.3 (Distance lower bound)**.** *Fix $0 < p < \alpha$. Every nonnegative continuous function, zero on Σ and harmonic in D, with*

$$\sup_{B_1} v = 1, \qquad v(x) \le \max(1, |x|^p), \tag{3.4}$$

satisfies

$$v(x) \ge a \operatorname{dist}(x, \Sigma)^q \quad (x \in D \cap B_1), \tag{3.5}$$

for some $a > 0$ independent of v.

Proof. Lemma 3.2 gives a common positive lower bound on $K_\eta = \{z \in \overline{B}_3 : \operatorname{dist}(z, \Sigma) \ge \eta\}$. For $x \in D \cap B_1$, let y be its nearest point on Σ, $d = |x - y|$, and $\zeta = (x - y)/d$. Convex projection gives $(x - y) \cdot (z - y) \le 0$ for $z \in \Sigma$. Taking $z = 0, 2y$ yields $(x - y) \cdot y = 0$. In particular $|y| \le 1$, $0 < d \le 1$, and $\Sigma \subset \{z : \zeta \cdot z \le 0\}$. Thus $B_1(y + \zeta) \subset D$, and its inner ball of radius $1 - \eta$ lies in K_η.

A sufficiently small fixed multiple $a\phi(\cdot - y - \zeta)$ is below v on that inner ball and outside the unit ball. Comparison on the outer collar, using (2.6), makes the same order hold everywhere. Evaluation at x gives

$$v(x) \geq a(1 - |x - y - \zeta|^2)^q = a(2d - d^2)^q \geq ad^q.$$

□

For a nonnegative function define its polynomial growth norm by

$$\|u\|_{p,*} = \sup_{R\geq 1} R^{-p} U_u(R), \qquad U_u(R) = \sup_{B_R} u. \tag{3.6}$$

This measures the whole tail, including the directions charged by μ.

Proposition 3.4 (Growth controlled by one value)**.** *Fix $q < p < \alpha$ and $x_* \in D \cap B_{1/2}$. For every positive continuous harmonic profile in D, zero on Σ, whose actual polynomial growth exponent is strictly below p,*

$$\|u\|_{p,*} \leq Cu(x_*). \tag{3.7}$$

The constant is independent of the profile and its initial growth constant.

Proof. Write $U = U_u$. Continuity makes $U(R)$ continuous, and the strict growth margin makes $U(R)/R^p \to 0$. Its positive supremum is therefore attained at some $R_0 \geq 1$. The function

$$v(x) = \frac{u(R_0 x)}{U(R_0)}$$

satisfies (3.4): the maximizing property gives its bound for $|x| \geq 1$, while $U(R_0)$ gives the bound inside the unit ball. The stable scaling identity $L(u(R\,\cdot))(x) = R^\alpha (Lu)(Rx)$ and the cone property preserve harmonicity and the zero set.

Apply Lemma 3.3 at x_*/R_0. Since the cone distance scales linearly,

$$u(x_*) \geq aU(R_0) R_0^{-q} \operatorname{dist}(x_*, \Sigma)^q.$$

Consequently

$$\|u\|_{p,*} = \frac{U(R_0)}{R_0^p} \leq a^{-1} \operatorname{dist}(x_*, \Sigma)^{-q} R_0^{q-p} u(x_*) \leq Cu(x_*),$$

because $p > q$ and $R_0 \geq 1$. □

4 Comparison and proportionality

Let β_1, β_2 be the growth exponents of the original profiles w_1, w_2 in Theorem 1.1. Choose once and for all

$$\max(q, \beta_1, \beta_2) < p < \alpha, \qquad x_* \in D \cap B_{1/2}. \tag{4.1}$$

All profiles considered below have actual growth exponent below this same p. Normalize two of them by $u(x_*) = v(x_*) = 1$. Proposition 3.4 and Lemma 3.2 give

$$u(x) + v(x) \leq C(1 + |x|)^p, \qquad \inf_K v \geq c_K > 0 \quad (K \Subset D), \tag{4.2}$$

with constants uniform over such normalized profiles.

The next comparison uses a bump outside the ball of interest. Its forcing comes from an angular cap of positive measure.

Lemma 4.1 (Annular forcing)**.** *There is $\eta_0 \in C_c^\infty(D)$, $0 \le \eta_0 \le 1$, supported in an annulus about radius $3/4$, outside $B_{2/3}$, such that*

$$L\eta_0 \le -c_0 < 0 \quad \text{in } D \cap B_{2/3}. \tag{4.3}$$

Proof. For $z \in D \cap B_{2/3}$, let ζ be the outward unit normal from its projection onto Σ. The projection identities in the proof of Lemma 3.3 give $\Sigma \subset \{\zeta \cdot x \le 0\}$ and $\zeta \cdot z = \operatorname{dist}(z, \Sigma) > 0$. By (2.3) and evenness,

$$\mu\{\theta : \zeta \cdot \theta \ge \delta\} \ge m, \qquad \delta = (\lambda/(2\Lambda))^{1/\alpha}, \quad m = \lambda/4.$$

Choose $2/3 < a_- < a_+ < 4/5$, both near $3/4$. Every ray $z + r\theta$ with $\zeta \cdot \theta \ge \delta$ crosses this radial band over an interval with

$$a_- - 2/3 \le r \le a_+ + 2/3, \qquad r_+ - r_- \ge a_+ - a_-.$$

The length bound follows from the 1-Lipschitz property of the radial norm. Throughout the crossing, $\operatorname{dist}(z + r\theta, \Sigma) \ge \zeta \cdot (z + r\theta) \ge r\delta$. Take $0 < \rho < \delta(a_- - 2/3)$, and choose a smooth bump equal to one on the compact set $\{a_- \le |x| \le a_+,\ \operatorname{dist}(x, \Sigma) \ge \rho\}$, supported in a slightly wider annulus outside $B_{2/3}$ and in $\{\operatorname{dist}(x, \Sigma) > \rho/2\}$. At every z in the tested ball its value is zero, and hence

$$-L\eta_0(z) = \int \eta_0(z + h)\nu(dh) \ge m(a_+ - a_-)(a_+ + 2/3)^{-1-\alpha} > 0.$$

□

Proposition 4.2 (Comparison of normalized profiles)**.** *For the fixed p, x_* in (4.1), there is $C_1 \ge 1$ such that every normalized pair as above satisfies*

$$C_1^{-1} u \le v \le C_1 u \quad \text{on } B_{1/2}. \tag{4.4}$$

Proof. Take smooth cutoffs $0 \le \chi, \xi \le 1$, with $\chi = 1$ on $B_{3/4}$, $\chi = 0$ outside $B_{4/5}$, $\xi = 1$ on $B_{1/2}$, and $\xi = 0$ outside $B_{2/3}$. Set

$$f = \chi u + A(\xi - 1) + B\eta_0.$$

In $D \cap B_{2/3}$, the cutoff identity in the viscosity sense gives

$$L(\chi u)(z) = \int (1 - \chi(z + h))u(z + h)\nu(dh) \le C.$$

The uniform bound follows from (4.2), the positive separation of $B_{2/3}$ from the support of $1 - \chi$, and $p < \alpha$. Also $|L\xi| \le C$. First take $A > \sup_{B_1} u$, then choose B large enough that (4.3) gives $Lf \le -1$ in the tested domain. Both constants can be chosen uniformly over the normalized family.

For $|z| \ge 2/3$, $\chi u - A \le 0$, so $f \le B\eta_0$. On $\operatorname{supp} \eta_0 \Subset D$, (4.2) supplies a common positive lower bound for v. Thus $f \le C_1 v$ outside the tested ball. On Σ, $f \le 0 = C_1 v$. Lemma 2.2 on $D \cap B_{2/3}$ gives $f \le C_1 v$ everywhere. In $B_{1/2}$, $f = u$. Interchanging u, v proves the claim. □

This comparison concerns entire cone profiles with controlled polynomial growth. The control at infinity in Proposition 3.4 is what makes its constant uniform over normalized profiles.

Proof of Theorem 1.1. For any two positive profiles u, v and $R \ge 1$, apply Proposition 4.2 to

$$u_R(x) = \frac{u(Rx)}{u(Rx_*)}, \qquad v_R(x) = \frac{v(Rx)}{v(Rx_*)}.$$

These remain in the same class. Evaluation at x_*/R bounds $u(Rx_*)/v(Rx_*)$ between C_1^{-1} and C_1 times $u(x_*)/v(x_*)$. For any fixed $x \in D$, choose $R > 2|x|$ and evaluate once more at x/R. We obtain the global comparison

$$C_1^{-2}\frac{u(x_*)}{v(x_*)} \le \frac{u(x)}{v(x)} \le C_1^2\frac{u(x_*)}{v(x_*)} \qquad (x \in D). \tag{4.5}$$

Let $\kappa = \inf_D u/v > 0$, and put $h = u - \kappa v$. By Lemma 2.2, h is harmonic in D. It is continuous, nonnegative, zero on Σ, and has growth exponent at most $\max(\beta_1, \beta_2) < p$. If it is nonzero, the strong maximum principle and connectedness proved in Lemma 3.2 make it positive throughout D. Apply (4.5) to h, v. This forces $\inf_D h/v > 0$, contradicting $\inf_D h/v = \inf_D(u/v - \kappa) = 0$. Hence $h = 0$, and $u = \kappa v$ everywhere. The choices in (4.1) are always possible because each original exponent is strictly less than α. They impose no further condition on the theorem. □

AI use. OpenAI Codex was used to develop the argument, check cited source statements, draft the manuscript, and perform internal mathematical and editorial review. The manuscript is submitted for independent mathematical scrutiny.

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